

Algorithm 1 Theory-constrained batch causal-effect estimation over 24Model factors

Require: Canonical HEC-KG $G^* = (V^*, E^*)$; accident cases $\mathcal{D}$; prior graph G_{24} ; treatment families Λ_{batch} ; $s_{\min} = 20$; $k_{\max} = 20$; $z_{\max} = 12$; $B = 500$.

Ensure: Effect records $\mathcal{A}$ and refutation records $\mathcal{R}$ with review status.

1: Build the causal dataset and select candidate treatments

$(\mathbf{X}, \phi, Y) \leftarrow \text{BuildCausalDataset}(V^*, E^*, \mathcal{D})$; $C_{\text{sel}} \leftarrow \text{Top}_{k_{\max}}(\Lambda_{\text{batch}}, \text{isLayer} \downarrow, \text{supp} \downarrow)$ Eqs. (14)-(16)

2: Screen treatment quality and select admissible common causes

If T or Y lacks variation, an arm has $< s_{\min}$ cases, or a $T \times Y$ cell is empty:

record $\hat{\tau}_T = \text{NA}$, set $\text{needs_review} = \text{True}$, and continue.

$\mathbf{Z}_T \leftarrow \text{SelectCommonCauses}(G_{24}, T, \phi)$; $|\mathbf{Z}_T| \leq z_{\max}$; exclude own-layer variables, descendants, and AccidentType. Eq. (18)

3: Identify, estimate, and quantify uncertainty

$G_T \leftarrow \text{BuildTreatmentGraph}(T, Y, \mathbf{Z}_T)$; $\mathcal{M}_T \leftarrow \text{CausalModel}(\mathbf{X}, T, Y, G_T)$

$\delta_T \leftarrow \mathcal{M}_T.\text{identify_effect}()$; $\hat{\tau}_T \leftarrow \mathcal{M}_T.\text{estimate_effect}(\delta_T, \text{backdoor.linear_regression})$ Eqs. (19)-(21)

$(\text{RD}_T, \text{CI}_{95\%}, \hat{\text{se}}_{\text{boot}}, p_T) \leftarrow \text{CaseBootstrap}(\mathbf{X}, T, Y, \mathbf{Z}_T, B, \text{seed} = 2026)$ Eq. (22)

4: Run refutation diagnostics and return auditable records

$\mathcal{R}_T \leftarrow \{\text{random common cause, placebo treatment}\}$; flag failed diagnostics; append $(T, \mathbf{Z}_T, \hat{\tau}_T, \text{RD}_T, \text{CI}_{95\%}, p_T)$; return $\mathcal{A}, \mathcal{R}$.